

Project Initialization and Planning Phase

Date	07 July 2026
Team ID	xxxxxxx
Project Title	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The **Voice-Based Concept Understanding Analyser (VBCUA)** is an AI-powered web application designed to evaluate students' conceptual understanding through spoken responses. Traditional assessments often emphasize memorization and require significant manual effort for evaluation. VBCUA leverages speech recognition and Natural Language Processing (NLP) techniques to automatically analyse spoken answers, measure conceptual understanding, and generate detailed performance reports. This solution improves assessment accuracy, reduces faculty workload, and provides immediate, objective feedback to students.

Project Overview	
Objective	The primary objective of this project is to develop an intelligent voice-based assessment system that evaluates students' conceptual understanding by analysing their spoken explanations. The system aims to provide accurate scoring, meaningful feedback, and automated report generation using Artificial Intelligence and Natural Language Processing techniques.
Scope	The project focuses on developing a Streamlit-based web application that records or accepts students' voice responses, converts speech into text, evaluates conceptual understanding using semantic similarity analysis, generates performance scores, and produces downloadable PDF reports. The solution is intended for educational institutions,

	faculty members, and students to support efficient and objective learning assessment
Problem Statement	
Description	Traditional written examinations and manual viva assessments often fail to accurately evaluate students' conceptual understanding. Manual evaluation is time-consuming, subjective, and difficult to scale for large numbers of students. Additionally, students rarely receive immediate and personalized feedback on their spoken explanations.
Impact	Addressing these challenges will improve assessment accuracy, reduce evaluation time, provide consistent scoring, enhance student learning through instant feedback, and support educational institutions in conducting efficient concept-based assessments
Proposed Solution	
Approach	Develop an AI-powered web application that records or accepts students' spoken responses, converts speech into text using speech recognition technology, evaluates conceptual understanding using semantic similarity techniques, calculates performance scores, and generates comprehensive feedback reports through an interactive Stream lit interface.
Key Features	➤ Voice recording and audio file upload support. ➤ Automatic Speech-to-Text conversion. ➤ AI-based conceptual understanding evaluation using semantic similarity. ➤ Performance scoring and detailed feedback generation. ➤ Downloadable PDF assessment reports. ➤ Interactive and user-friendly Stream lit interface.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		

Computing Resources	Processor	Intel Core i5 (or higher)
Memory	RAM	8 GB
Storage	Disk space for data, models, and logs	20 GB
Software		
Programming Language	Backend Development	Python 3.11
Frameworks	Web Application	Stream lit
Libraries	AI & Data Processing	OpenAI Whisper, Sentence-BERT, Librosa, PyDub, Pandas, NumPy, ReportLab
Development Environment	IDE	Visual Studio
Data		
Input Data	Voice Responses	Recorded Speech (.wav/.mp3)
Reference Data	Conceptual Answers	Text-based Reference Answers
Output Data	Evaluation Reports	Scores, Feedback, PDF Reports